have this nasty property. They much better able to preserve the original distribution. However, they introduce an extra source of variability.

Another point to take into consideration is the effect of imputation on relationship between variables. Several imputation techniques cause estimates of covariances and correlations to be biased. The estimates produce too low values. For more information, see e.g. Kalton and Kasprzyk (1986).

Research for new imputation techniques is in progress. An example is the multiple imputation technique proposed by Rubin (1979). This technique computes a set of imputed values for each missing value. This results in a number of imputed data sets. Inference is based on the distribution of estimates obtained by computing the estimate for each data set. This approach has attractive theoretical properties, but may not be so easy to implement in practical situations. An overview of the state of the art in nonresponse research can be found in Groves et al. (2001), Chen and Haziza (2019), and Scheaf et al. (2023).

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Nonresponse in Web Surveys

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Introduction

Reaching and obtaining responses from target individuals is one of the crucial determinants of data quality in web surveys, just as in any other survey mode. Lack of interviewers, computerized and visually presented survey questions and data transmission through the Internet lay the foundation for many advantages of web surveys, but may adversely affect the success in reaching target persons and their willingness or ability to participate in the survey.

Declining response rates have challenged the survey industry over the last decades (Dillman et al. 2014). Failure to contact target persons, refusals and inability to complete the questionnaire for technical or personal reasons all reduce the success of obtaining information from the target population. Response rates achieved by web surveys are highly variable, but are commonly perceived as lower compared to traditional survey modes, such as face-to-face, telephone and mail surveys. Two meta-analyses of experimental studies (Daikeler et al. 2020; Lozar Manfreda et al. 2008) indicated that web surveys on average yield about 12 percentage points lower response rate than other survey modes. However, such comparisons often do not separate the effect of the web mode and recruitment method, where web surveys are often used with inexpensive and less effective recruitment approaches. There is little evidence that web survey would perform worse than another survey mode in terms of cooperation if the same recruitment strategy was used in both modes.

This section focuses on unit nonresponse as a failure to obtain any or sufficiently complete data from a target person or other unit of interest (e.g. a household or an enterprise). This can occur due to individuals not participating in the survey at all, terminating the participation prematurely or finishing the questionnaire with too high item nonresponse for their answers to be useful. While web surveys do not tend to produce significantly higher item nonresponse than other survey modes (Čehovin et al. 2022), breakoffs occur more frequently (Peytchev 2011; Tourangeau et al. 2013). Variations in breakoffs in web surveys reported by various studies are profound, ranging from a fraction of percentage to more than half of those opening the web survey questionnaire (Liu and Wronski 2018; Revilla 2017; Vehovar and Čehovin 2014). Early breakoffs can therefore be a significant contributor to unit nonresponse in web surveys and need to be considered as part of nonresponse discussion.

Factors Affecting Web Survey Nonresponse

External reasons for generally low response rates in web surveys were identified already by Vehovar et al. (2002) and further elaborated in Callegaro et al. (2015). They stem from the socio-technical context in which surveying takes place, personal characteristics of target individuals, survey design and information-communication infrastructure.

As regards **the socio-technical context**, web surveys enabled low-cost survey data collection too all interested individuals. Millions of “do it yourself” surveys recruit all segments of the population, increasing the problem of *survey fatigue due to oversurveying* and difficulties distinguishing between serious research requests, low-quality surveys and marketing campaigns (Callegaro et al. 2015).

Other specifics of the Internet can contribute to lower participation in web surveys as well. Trust in sweb urvey invitations may be lower because of *privacy concerns, fraudulent request, and difficulties establishing trust and legitimacy due to lack of personal contact* (Fan and Yan 2010; Keusch 2015). Further challenges occur when invitations are sent by e-mail due to *frequent invalid addresses, misidentification of invitation messages as spam, irregular e-mail checking and overlooked communication* (Callegaro et al. 2015). The situation is even more problematic with publicly posted invitations through online banners, ads or intercept survey invitations, which are often clicked only by a small fraction of visitors.

Another important set of factors arise from **personal characteristics** of individuals targeted by the survey. Topic salience and interest, psychological traits (e.g. sense of obligation, need for cognition and cognitive abilities), attitudes towards the survey sponsor, IT skills and trust in technology are some examples of characteristics that may determine the individual decision (or ability) to complete the questionnaire online (Callegaro et al. 2015; Keusch 2015). The influence of these characteristics on data quality is especially damaging if they are correlated with the survey topic. This extends to online panel surveys where motivation to participate in the panel may be related to specific attitudes, personality traits or other characteristics. For example, studies found that panelists may be heavier internet and media users, more interested in technology and politics, more sensitive to environmental issues, less traditional and less concerned about privacy than general population members (Callegaro et al. 2014).

The **information-communication infrastructure** raises the issue of Internet *noncoverage,* which is often difficult to separate from nonresponse. Although the proportion of population using the Internet is almost 100% in some countries (like Iceland, Saudi Arabia and Lichtenstein), this is not the case in many other, even most developed economies (World Bank 2023). Some population segments, such as elderly, lower educated and persons with lower socio-economic status are less likely to use the Internet (Eurostat 2023) and may therefore be underrepresented in general population web surveys despite overall high Internet penetration. This issue is even more pronounced in developing and least developed countries, several of which have less than a third of population using the Internet (World Bank 2023). Internet penetration therefore remains a significant country-level factor of web survey participation (Daikeler et al. 2022).

These external factors can significantly affect data quality in terms of representation, but researchers cannot do much to influence them. On the other hand, the **design and implementation of a web survey** is under a direct control of researchers and can be used to their benefits. Comprehensive guidelines for effective recruitment of target persons have been proposed to maximize the propensity of an individual to participate in a web survey (e.g. Dillman et al. 2014).

A **recruitment strategy** needs to be tailored to specifics of the survey and the target population. General recommendations include multiple contact attempts to reach target units, relying on more trustworthy contact modes (e.g. mail in addition to e-mail invitations), assuring motivational content of survey request, using alternative survey modes for non-respondents, and incentives (Callegaro et al. 2015; Dillman et al. 2014; Wu et al. 2022). While findings regarding the effectiveness of the latter are inconsistent (Wu et al. 2022), pre-paid monetary tokens of appreciation generally outperform other incentive types, like post-paid and lottery-type incentives (Callegaro et al. 2015; Dillman et al. 2014).

Questionnaire design also guides individual's decision to participate and provide (sufficiently) complete answers. Questionnaire length, complexity and visual layout are significant contributors to survey breakoffs and need to be optimized to reduce the burden of respondents (Liu and Wronski 2018; Tourangeau et al. 2013). This is becoming even more critical with the increasing use of mobile phones for questionnaire completion, which have significantly higher breakoff rates (Mavletova et al. 2015; Steinbrecher et al. 2015), calling for careful and methodologically sound adaptation of web questionnaires (Antoun et al. 2018; Vehovar et al. 2022).

The researchers have some flexibility also in relation to **survey sponsor** (e.g involving an highly reputable sponsor), **survey topic** (e.g. combining or exposing salient topics or aspects), **mode**, **timing and frequency of contacts**, length of the **fielding period**, and **invitation format** (personalization, motivation and tone of request, visual design). All these components can significantly improve response rates (Callegaro et al. 2015).

Web Survey Nonresponse Assessment

The **achieved survey participation** can be measured by various methods that at minimum require distinguishing between respondents, nonrespondents, eligible and ineligible units (AAPOR 2016). In web surveys, these outcomes are often difficult to establish for each target unit. It is generally only possible with list-based web surveys (Callegaro et al. 2015), where target units from a sample are provided with a unique questionnaire access code that allows tracking survey outcomes at the individual level. This is often not feasible due to lack of comprehensive lists of target units (e.g. no e-mail lists of the general population exists, lists of different target populations are often not covering all or not including complete contact information etc.) or unrestricted access to the web questionnaire. When invitations are publicly posted, for example in online forums, magazines, or specific websites, or when recruiting for nonprobability online panels (Callegaro et al. 2014), the exact number of persons exposed to the invitation is often unknown and nonresponse cannot be assessed at all.

Frequent breakoffs further complicate assessment of response rates. The researcher needs to decide on a criteria of questionnaire completeness to distinguish between complete respondents, partial respondents and nonrespondents. One example (AAPOR 2016) is to consider cases with less than 50% of applicable questions answered as non-respondents, those with 50–80% as partials and those with more than 80% as complete respondents.

Some commonly used **survey participation metrics** that can be calculated for list-based web surveys based on the established survey outcome statuses include (AAPOR 2016):

- **Response rate** is the number of respondents (complete-only or complete and partial respondents) divided by the number of all eligible units in the sample. In online panels, the total response rate of a particular panel survey is the product of response rates across several stages: panel recruitment (recruitment rate), profiling (profile rate) and the individual survey (completion rate) (AAPOR 2016).
- **Participation rate** is an alternative participation indicator to response rate for panel surveys in which the number of eligible units is not known for each stage of panel recruitment, like with most non-probability online panels (Callegaro et al. 2014). It is calculated as the number of (complete or partial) respondents to a particular panel survey divided by the number of panelists invited to participate in that particular survey (AAPOR 2016; Baker et al. 2013).
- **Contact rate** is the number of successfully contacted units divided by the number of all eligible units in the sample. Because the lack of feedback from nonresponding units makes eligibility and contact success difficult to establish in many web surveys, it is often more feasible to calculate the **absorption rate** (Vehovar et al. 2002)—the ratio between the number of invitations considered to be successfully delivered (e.g. not returned as undelivered) and the number of all invitations sent by e-mail, mail or any other mode.
- **Cooperation rate** is the number of (complete or partial) respondents divided by the number of all successfully contacted units. In web surveys, this can be primarily used as an indicator of solicitation success but is only feasible if contact success can be reliably determined.

  For list-based web surveys an approximate measure of the cooperation rate can be calculated by assuming that all units with absorbed invitations were also contacted.

Many other indicators can be derived to evaluate specific aspects of survey participation (like proportion of breakoffs among all who accessed the questionnaire) or to address specific survey designs (like mixed-mode surveys).

Consequences of Web Survey Nonresponse and Adjustment

A concerning potential consequence of nonresponse is the **nonresponse bias**. The amount of the bias depends on the proportion of nonrespondents (i.e. the nonresponse rate) and the magnitude of the difference between respondents and nonrespondents on a particular survey variable (Groves 2006). While generally lower survey participation can be regarded as one of the main downsides of web surveys, it therefore does not necessarily translate into lower data quality due to nonresponse bias. Nonresponse problem assessment for each survey needs to consider the level of nonresponse as well as potential differences between respondents and nonrespondents.

The **difference between respondents and nonrespondents** on characteristics of interest is more challenging to assess than the level of nonresponse. Little of relevance is usually known about those who did not participate in the survey. When some background information is available (e.g. from the sampling frame, other surveys or census data), the structure of the obtained survey sample and these benchmark data can be compared (e.g. Petrovčič et al. 2016). This is usually limited to very basic characteristics (like gender, age and area of residence) and the relevance for nonresponse bias assessment depends on the strength of the correlation between these characteristics and survey variables of interest. The matter is further complicated as different survey variables are likely differently affected by the nonresponse bias.

Another approach is to **compare early and late respondents**, i.e. those that responded to the survey request faster to those that required more solicitation effort. By assuming that the latter are more like nonrespondents than the former, some indication of the direction and magnitude of the nonresponse bias may be obtained. This approach can be used for all survey variables, but the accuracy of the underlying assumption is difficult to verify for a particular survey.

In addition to using strategies for increasing and methods for monitoring survey participation, another set of measures attempts to improve the estimates through **post-survey adjustment** techniques, like post-stratification weighting and propensity score adjustments. Their effectiveness strongly depends on correlations between survey nonresponse, available control variables with known values for respondents and nonrespondents, and target survey variables (Callegaro et al. 2014; Cornesse et al. 2020; Schonlau et al. 2009). While post-survey adjustments are valuable and generally recommended tool to improve accuracy of survey estimates, they should therefore not be relied on as an alternative for carefully designed recruitment strategy and nonresponse reduction measures.

Conclusion

Trends of declining response rates will likely continue to affect the survey industry, including web surveys, and further development of measures to increase response rates will be crucial to assure successful survey recruitment. At the same time researchers need to respond to cost-reduction pressures, limiting the resources available for survey implementation. The recruitment strategy therefore needs to be tailored to achieve the balance between the estimated risk of nonresponse bias, required accuracy of survey estimates and practical as well as budgetary concerns.

The revolution that web surveys brought into survey research is fueled primarily by their cost efficiency in addition to data collection speed and convenience. Response rate alone is therefore largely incomplete survey quality benchmark. The inclusion and observation of corresponding nonresponse bias is already an important extension, but for a comprehensive evaluation, it is essential to include survey costs (Vehovar et al. 2010; Vehovar and Beullens 2018). Namely, if all costs and errors are considered, even five times lower response rate in a web survey can outperform the face-to-face survey alternative, as shown on a case of Slovenian travel survey (Bučar 2022). Similar can be observed with the European Social Survey, which relaxed its requirement for target response rate of at least 70%, started introducing web mode as part of mixed-mode design and even explores probability web panels where substantially lower response rates can be expected (ESS ERIC 2021; Maslovskaya and Lugtig 2022). At the extreme end of this trend there is the proliferation of nonprobability web panels, where response rates cannot even be assessed anymore. However, such panels have become an important research tool and started replacing some surveys based on probability samples not only in market research, but also in academic sector and even in official statistics (Vehovar et al. 2023).

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Nonsampling Errors in Survey Research

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The term "Nonsampling Errors" has long been employed to refer to all the potential biases and variance in a sample survey, other than the error that stems from sampling (which is the inherent variability of estimates from sample to sample).

Survey statisticians' concerns about sampling error generally outweighed their attention to "survey error" until the 1940s when Deming (1944) rightly called attention to the fact that there are multiple types of potentially serious errors inherent in many surveys that should be of concern to the statistical community. Then, in the 1950s, Hansen and Hurwitz (1958) addressed the serious and growing problem of survey nonresponse and its associated errors. But it was not until the late 1980s that Groves (1989) provided a very deep dive into both nonsampling errors and sampling errors in surveys, in his *Survey Errors and Survey Costs*. This seminal work, fueled the current era of considerable attention being paid to Total Survey Error (TSE) in which the survey research community now engages (Lavrakas 2013).

Nonsampling Error includes five of the error categories in the TSE framework, as shown in Fig. 1 (adapted from Groves et al. 2009). Those are the Errors of Representation that are related to Coverage, Nonresponse, and Adjustment; and the Errors of Measurement that are related to Specification, Response – including errors associated with the Questionnaire, the Respondents, the Interviewers, and the Data Collection Mode(s) – and Processing errors.

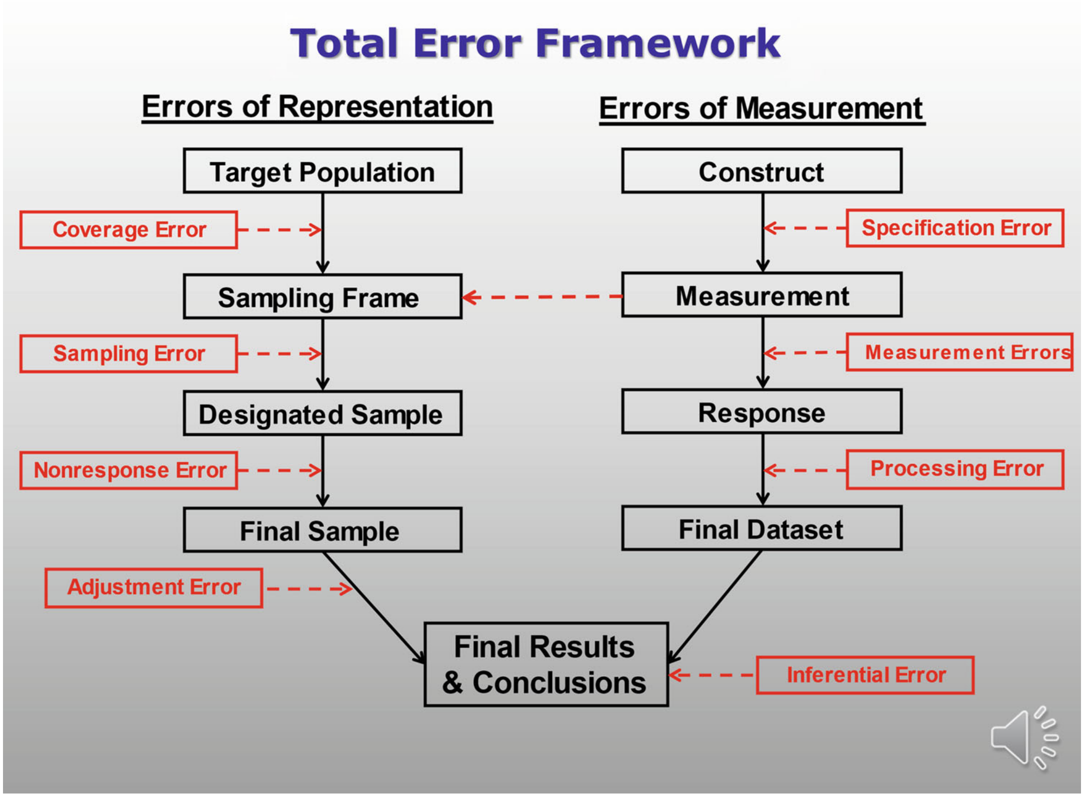
In this entry, these types of errors will be discussed separately, but the reader should note that in practice, the issues overlap and interact. For example, a poorly worded question that produces confusion with a self-administered mail-back questionnaire may in contrast cause no error when deployed via an in-person or telephone interviewer. In addition, a brief entry such as this cannot address all possible nonsampling errors that are discussed in the literature. Please see the references at the end of the entry for fuller treatments; see also the entry on ► "Total Survey Error".

Nonsampling Errors of Representation

Coverage/Noncoverage and Associated Coverage Errors

Many survey researchers fail to properly identify their target population. This failure can create problems that likely are ignored, or even go undetected, by the researchers in determining how well they have covered that population with their sampling frame(s). Noncoverage is a problem that occurs when a portion of the target population is not included ("covered") in the survey's sampling frame(s). This coverage problem can lead to nonnegligible coverage error if those, as a group, who were not included in the frame(s), and thereby who could not have been sampled, would have provided different data for the survey's measures of interest, compared to the data that were gathered from those who were included and selected from the sampling frame(s).

Undercoverage occurs when members of the target population are missing from the frame from which the sample is drawn. Error (bias) results when the undercoverage is correlated with the type of people who, as a group, would have provided different data compared to those who were adequately covered by the frame that was used. Early in the twentieth century, sample surveys in the United States were done via in-person interviewing or by mail because telephone penetration was so low as to make telephone numbers a frame that suffered from marked undercover-



Nonsampling Errors in Survey Research, Fig. 1 TSE Conceptualization of Survey Errors

age in representing the U.S. population; and the undercoverage was of a differential nature; i.e., it was not a random subset of the population that was without a telephone at home (Groves and Kahn 1979); (e.g., the non-telephone population was of lower SES than the telephone population, thus biasing the sample on any SES variables and any correlated with SES). As telephone penetration grew close to 100% towards in the last quarter of the twentieth century, telephone interviewing became the method of choice for many governmental and non-governmental surveys. But as more and more households replaced their landlines with cell phones, methods using random digit dialing landline sampling began to suffer from serious undercoverage, and methods had to be devised to reach the sizable and constantly growing cell-phone only population. A turn toward "address-based sampling" (with 98%-plus coverage of residences in the U.S.) during the twenty-first century was one solution to these undercoverage problems.

Overcoverage problems occur when the frame includes duplicates and/or individuals who are not members of the target population, as when those who will not vote in an election are nevertheless included in the frame(s) and sampled to be interviewed about their voting intentions, and their answers included in estimates of candidates' popularity among likely voters.

Nonresponse and Associated Unit Nonresponse Errors

If a unit from the population that is initially designated to be in the sample cannot be reached, refuses to participate in the survey, or if there is a barrier to gathering data from the unit (e.g., language), then what is called "unit nonresponse" occurs. To avoid unit nonresponse, survey researchers use various recruitment strategies to gain respondent cooperation during the field period such as making repeated contact attempts, offering noncontingent and/or contingent incentives to participate, pledging confidentiality or even anonymity, leveraging the reputation of their institution to instill legitimacy/trust, etc.

Unit nonresponse *error* in the form of bias occurs to the extent that nonrespondents are different from the respondents in the data they would have provided, and hence estimates of the quantities of interest in the survey are biased. On occasion, when funding is sufficient, a subsample of nonrespondents is pursued with special efforts after the initial field period ends to try to estimate the nature and size of the nonresponse biases on key survey items that resulted during the original field period. What is learned via these so-called NRFU (nonresponse follow-up) studies, and via other methods for investigating nonresponse biases (Montaquila and Olsen 2012) can (and should) be used to inform weighting decisions about how to adjust the dataset to try to reduce these biases before the substantive analyses are performed.

When a survey is longitudinal, a so-called panel survey, in which the same respondents are questioned several times over a period of months or years, there is a special kind of unit nonresponse (attrition) that occurs when a respondent who has given answers during previous rounds/waves of the survey cannot be contacted or refuses to answer in the current round. Prudent survey researchers who carry out longitudinal surveys put in place careful procedures to keep in touch with panel members between waves of data collection to help avoid this kind of panel nonresponse and the related error that may occur.

Weighting and Adjustment Errors

Apart from those adjustments due to the survey's sampling design (i.e., for design factors that have caused the final sample to deviate from a simple random sample), prudent survey researchers may adjust to try to reduce the possible biases associated with coverage errors and nonresponse errors.

In terms of dealing with coverage error, a basic approach is to compare the characteristics of one's initially designated sample to their population parameters. For a survey using addresses, this is done easily nowadays in many countries using auxiliary frame data (e.g., local block group variables gathered by a national census) that are provided by numerous sampling vendors that can be matched to all units in the survey's initial sample of addresses. It is not done as easily or usefully if the sampling unit is a telephone number or other point-of-contact (e.g., email addresses).

In dealing with nonresponse error, a basic approach is to compare the characteristics of the final (responding) sample to their population parameters. This is done easily nowadays in many countries with myriad auxiliary data (e.g., local block group variables gathered by a national census, and a myriad of other person-level and household-level "big data" sourced variables to which sampling vendors have access) that can be matched to units in the survey's final sample of addresses. Again, it is easily done for surveys that use addresses as the sampling unit, but not done as easily or usefully if the sampling unit is a telephone number or other point-of-contact.

Adjustment errors can result when (a) the data for the population parameters are faulty, (b) bad decisions are made about which characteristics are adjusted for, and/or (c) analytic carelessness occurs when the weight adjustments are created and/or applied.

Unfortunately, efforts to reduce one form of survey error, bias, tend to increase the amount of error created by inflating variance (i.e., increasing the survey's imprecision). Furthermore, adjustments to try to reduce nonsampling errors are often likely to increase sampling errors due to so-called design effects (*deff*), and as reflected in a reduction of a survey's effective sample size. (See entry on "Design Effects" elsewhere in this volume).

Nonsampling Errors of Measurement

Misspecification and Associated Specification Errors

Specification errors occur in the planning stage of a survey if researchers prepare an instrument that does not collect the data for all the necessary constructs and variables to meet the objectives of the study. If the researchers fail to adequately include measures in their questionnaire that are needed to validly measure the constructs they purported to have measured, those researchers have threatened the construct validity of their study and created